Milestone 5

Recommendation Engine & FastAPI API Prototype

1. Introduction

Milestone 5 focuses on transforming the trained machine learning model into a **usable decision-support system** by exposing predictions through an API and mapping them to **actionable business recommendations**.

While previous milestones concentrated on data ingestion, feature engineering, and model optimization, this milestone bridges the gap between **model output and real-world decision-making**.

The key objective of Milestone 5 is to **enable real-time credit risk scoring** and provide **clear loan approval or recovery recommendations** using a RESTful API built with FastAPI.

2. Objective of Milestone 5

The primary objectives of this milestone are:

- To design and implement a **recommendation engine** based on predicted loan default probability
- To expose the trained machine learning model through a **FastAPI-based REST API**
- To validate the system locally by testing real-time prediction requests
- To ensure recommendations are **logical, interpretable, and aligned with business use cases**

3. Recommendation Engine Design

3.1 Motivation

A raw probability score alone is not sufficient for business users such as loan officers or recovery agents. Therefore, a **rule-based recommendation layer** was designed on top of the model predictions to convert probabilities into **human-readable risk categories and actions**.

3.2 Risk Categorization Logic

The predicted probability of loan default is categorized into four risk levels:

Default Probability Range	Risk Level	Recommendation
< 0.30	Low Risk	Approve loan
0.30 – 0.55	Moderate Risk	Approve with monitoring
0.55 – 0.75	High Risk	Approve with higher interest rate
≥ 0.75	Very High Risk	Reject loan or demand collateral

This structured approach ensures:

- Consistent decision-making
- Transparent reasoning
- Alignment with financial risk management practices

3.3 Recommendation Output

For each borrower request, the system outputs:

- Default probability
- Risk category
- Suggested business action

This enables stakeholders to take **immediate and informed decisions**.

4. FastAPI-Based Serving Layer

4.1 Purpose of Using FastAPI

FastAPI was selected as the serving framework because it provides:

- High performance
- Automatic API documentation (Swagger UI)
- Built-in input validation
- Easy integration with machine learning models

4.2 API Architecture

The API performs the following steps:

1. Accepts borrower loan details as input
2. Validates inputs using a defined schema
3. Applies the same preprocessing (scaling) used during model training
4. Generates default probability using the trained model
5. Applies recommendation logic
6. Returns results as a structured JSON response

4.3 Input Parameters

The API accepts the following borrower attributes:

- LoanAmount
- InterestRate
- DTIRatio
- CreditScore
- MonthsEmployed

These features were selected based on their importance in credit risk modelling.

4.4 API Endpoints

Root Endpoint (`/`)

- Used to verify API health
- Returns a success message confirming that the API is running

Prediction Endpoint (`/predict`)

- Accepts borrower details via POST request
- Returns risk probability and recommendation

5. Input Validation & Robustness

To ensure reliability and data integrity, **input validation** was implemented using Pydantic models.
These guarantees:

- Correct data types
- Mandatory fields are provided
- Invalid or malformed requests are rejected before model inference

This step improves system robustness and aligns the API with industry standards.

6. Local Testing & Validation

6.1 API Execution

The API was executed locally using Uvicorn, confirming successful startup and runtime stability.

6.2 Swagger UI Testing

FastAPI built-in Swagger UI (`/docs`) was used to:

- Test the prediction endpoint interactively
- Validate correct request–response behaviour
- Verify recommendation logic across different risk scenarios

6.3 Validation Outcome

- API responded correctly for all valid test cases
- Risk categories changed logically with input variations
- Recommendations matched defined business rules

This confirms that the system satisfies all **Milestone 5 evaluation criteria**.

7. Key Outcomes of Milestone 5

- Successfully converted machine learning predictions into **actionable insights**
- Deployed a fully functional **FastAPI prototype**
- Enabled real-time credit risk scoring
- Ensured recommendation logic is transparent and explainable
- Established a foundation for frontend integration in the next milestone

8. Conclusion

Milestone 5 successfully delivers a **recommendation-driven API prototype** that operationalizes the CreditPathAI machine learning model.

By integrating a rule-based recommendation engine with a FastAPI serving layer, the system moves beyond offline modelling into a **deployable, decision-support solution**.

This milestone completes the backend inference phase and prepares the project for frontend development, dashboards, and stakeholder interaction in Milestone 6.